

COLLISION REPAIR INFORMATION BULLETIN



SUBJECT:	BODY SEAM AND JOINT SEALER APPLICATION		
BULLETIN NO.:	TS–TCI–CRIB–2025	DATE ISSUED:	May, 2013
MODELS:	All Toyota & Scion Models	SECTION:	Exterior

During the repair and restoration of a vehicle to its pre-accident condition, making the seam and joint sealer appear as they did during the assembly line process may be challenging due to the different types of materials and equipment used in the repair industry. This bulletin will summarize the different methods used to help restore the appearance of the sealers to pre-accident condition.

MATERIALS AND TOOLS USED:

A cartridge style body sealer tube used with either a manual gun or the air gun set for sealer application can be effectively used to apply sealer to hem joints and other areas of the vehicle.

Cartridge Body Sealer Tube



Air Gun Set for Body Sealer



APPLYING SEALER TO HEMMED PORTIONS (I.E.: ENGINE HOOD, DOOR SKINS, ETC.):

“Hemmed Joints” are joints with the edge of the panel bent over the edge of another panel such as the door skin or skin of the hood panel. There are two methods to apply sealer to hemming joints to help restore the factory appearance of the sealer:

- Use of a special beading nozzle attached to the end of a cartridge body sealer tube available by suppliers of sealer products.
- Cutting and shaping the cartridge body sealer tip to the shape and size of the factory sealer.

Refer to the following pages for instructions for both methods.

NOTE: Issue of this Collision Repair Information Bulletin by Toyota Canada Inc. is to ensure that proper repair information is available. It does not constitute an authorization for repair. PLEASE ROUTE THIS BULLETIN TO YOUR COLLISION REPAIR CENTER MANAGER AND COLLISION & REFINISH TECHNICIANS.



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Method 1: Use of Special Beading Nozzles

Many suppliers of body sealer cartridges make available special beading nozzles that attach to the end of the cartridge that provide a guiding edge and special outlet shape to match the factory sealer shape and size.

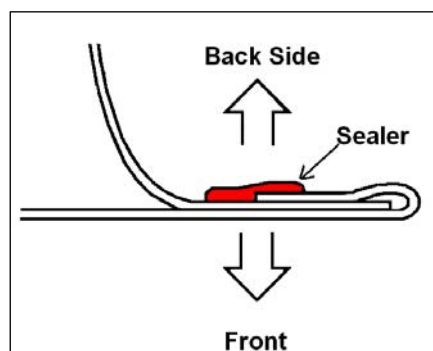
Toyota Canada recommends the Würth Special Beading Nozzle (Art. No. 891.628) that is available through the Würth Body Shop Products Program. These customized nozzles can produce bead sizes from 6 mm - 13 mm (1/4" - 1/2"). For more information, refer to the last page of this bulletin.



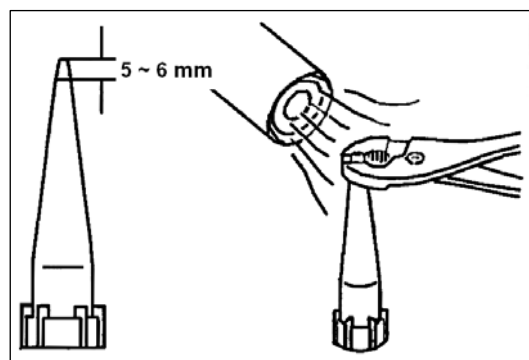
Method 2: Cutting and Shaping the Cartridge Body Sealer Tip

1. Preparing the Cartridge Nozzle

- a) Cartridge nozzles usually have a conical shape. Therefore, just cutting the end of the nozzle in the usual manner will cause the sealer to come out with a rounded shape. However, the sealer applied to hemmed portions at the factory is broad and flattened in shape as shown in the illustration to the right. In order to achieve this broad, flattened shape, it is necessary to prepare the nozzle as shown in the illustration on the next page.



- b) Cut 5-6 mm off the end of the nozzle with an appropriate cutter.
- c) Heat the nozzle with a heat gun and use a pair of pliers to flatten the end of the nozzle so the sealer will be applied in a flat smooth manner.



NOTICE: Excessive heat (290 °C or 100 °F) may melt the plastic nozzle.

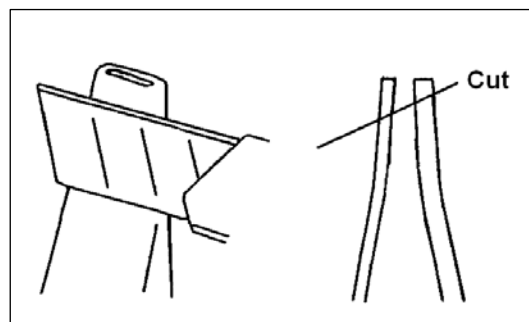
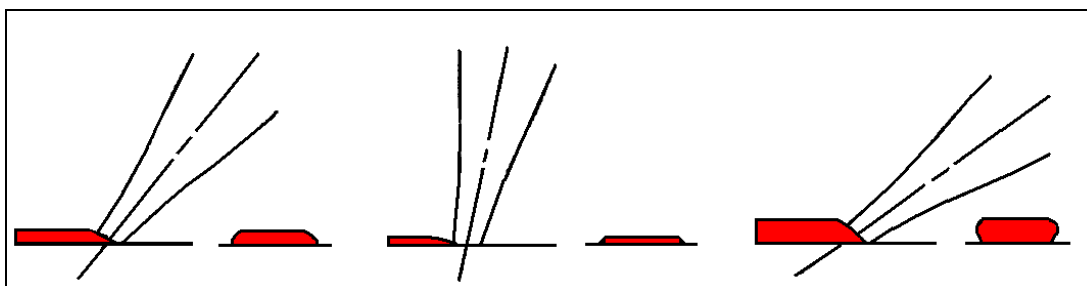
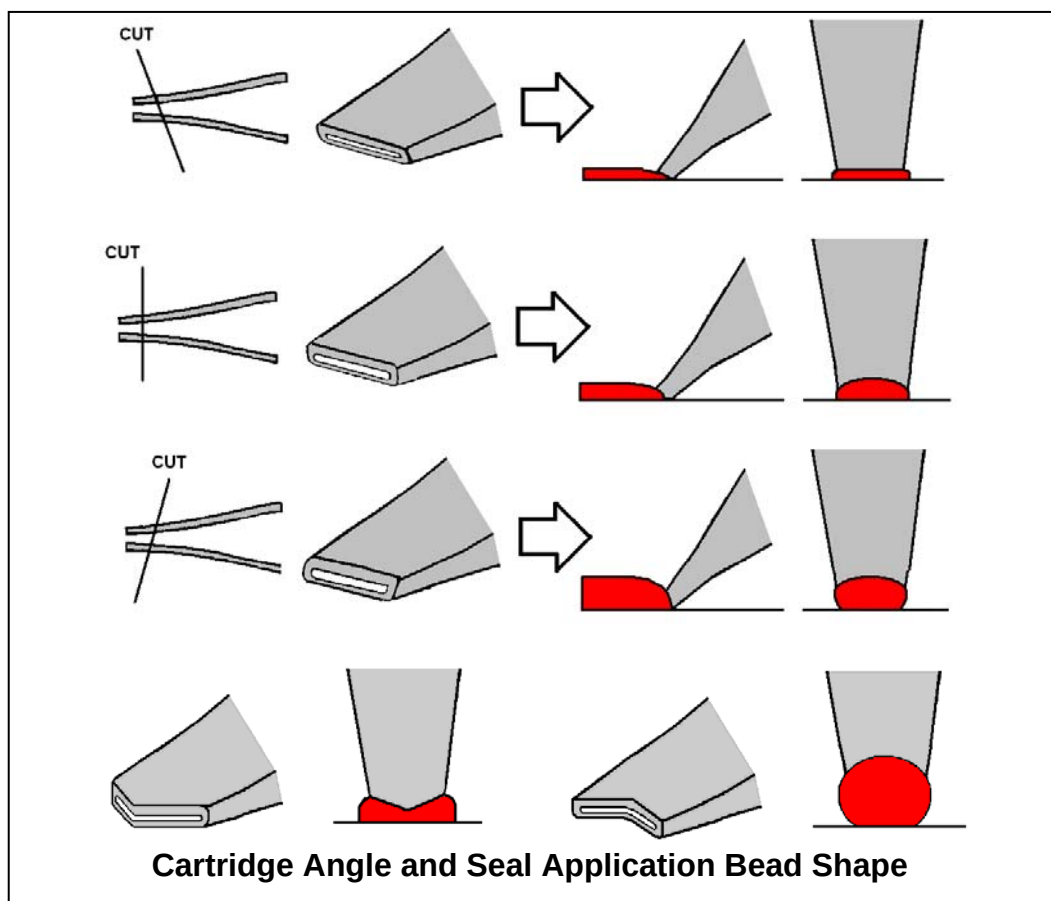


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- d) To shape the end of the nozzle, use a flat blade screw driver in the heated end of the nozzle.
- e) Cut the end of the nozzle at an angle with an appropriate cutter. The location on the nozzle where the cut is made will determine the width of the sealer.


REFERENCE: Nozzle Cutting and Sealer Application Bead Shapes




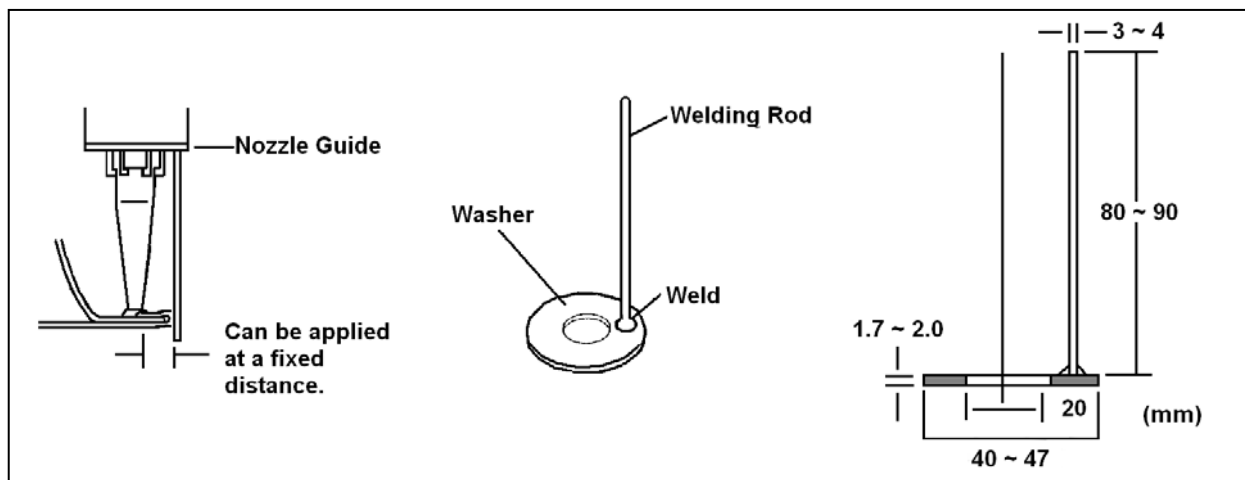
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2. Preparing a Nozzle Guide

- In order to apply sealer in a straight line along the hemmed portion without the bead “wandering around”, a guide may be necessary.
- A guide like the one shown below will allow sealer to be applied at a position which is at a fixed distance from the edge. The sealer can then be applied straight to help produce a good appearance and finish.
- To make a guide, follow the instructions and illustrations below using a simple washer and a welding rod.



NOTE: If the washer is too thick or thin, the nozzle will not tighten fully and may move back and forth during use. Use an appropriate size washer. If the proper size washer cannot be located, fasten two washers together and adjust the nozzle end by cutting it slightly.

3. Application Procedures :

- Degrease the area where the sealer will be applied. Be certain all wax and grease remover has been eliminated from the surface before applying the sealer. Do not apply sealer to bare metal - Always apply an epoxy primer to bare metal and new body panels.
- Use the nozzle guide to apply an even bead of sealer to edge of the panel. If the edge of the guide does not allow the sealer to be applied in the proper position, bend the guide rod as necessary.
- Apply a small amount of sealer to a test panel to check for correct amount and alignment of sealer being applied.

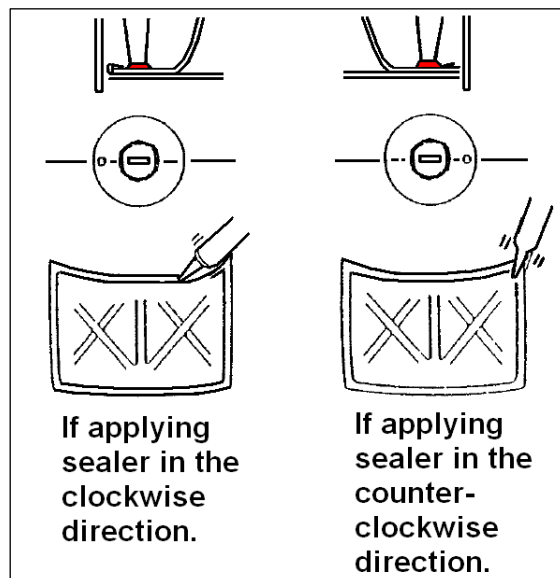


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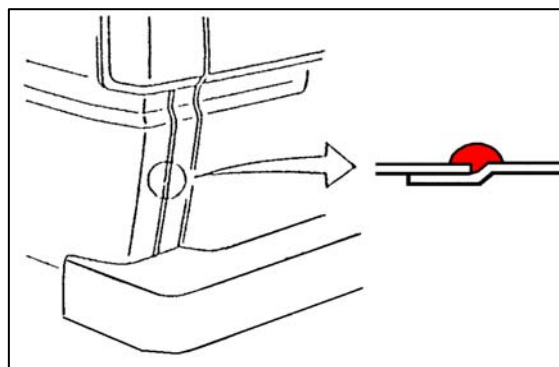
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- d) Apply a necessary amount of sealer to properly seal the panel from rust and corrosive agents following the edge in a clockwise or counter-clockwise direction as shown in the illustration to the right.



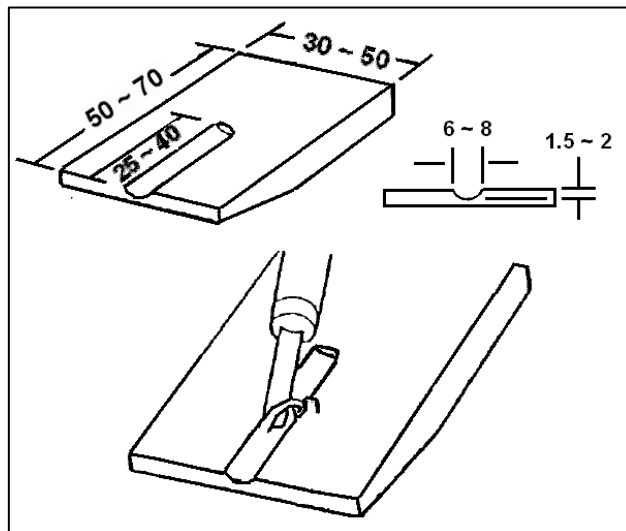
4. Applying Sealer to Rear Body Panels (exposed raised sealer joints):

In some cases, certain vehicles use an exposed raised sealer joint. To apply professional looking sealer application, the use of a shaping spatula may help the finished application of sealer appear as it did on the original joint.



Making a Shaping Spatula

- a) To make a shaping spatula, cut and shape the spatula as shown in the illustration to the right. Materials that can be used for the spatula are a hard sanding block or a piece of hard flat plastic. After the initial cutting of the groove has been made with an engraving tool, it will be necessary to sand the groove smooth with a fine grit sandpaper. This can be performed by folding the sandpaper into a U-shape. The depth and width of the U-shaped groove will differ depending on the vehicle and model being repaired.



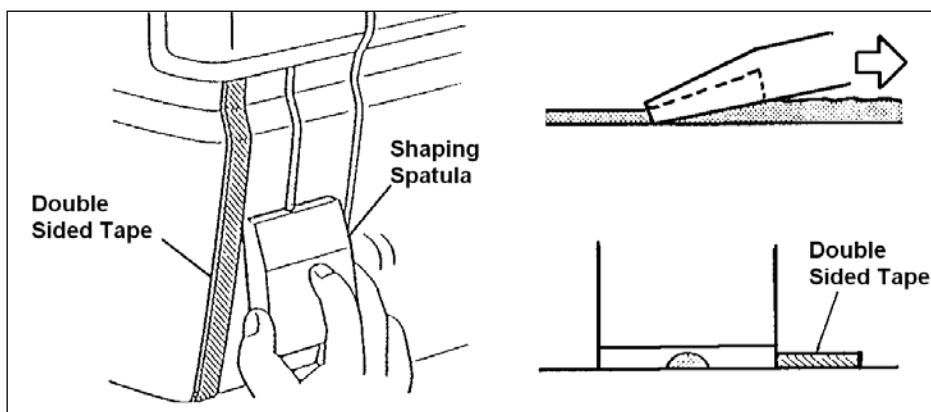


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- b) Degrease the area where the sealer will be applied. Be certain all the wax and grease remover has been eliminated from the surface before applying the sealer. Do not apply sealer to bare metal - Always apply an epoxy primer to bare metal and new body panels.
- c) Position the U-shaped groove of the shaping spatula in place on the portion where the sealer will be applied and affix thick double sided tape along the side of the tool. This double sided tape acts as a guide as the shaping spatula is moved along the panel.
- d) Apply sealer to the panel joint. Use the appropriate round bead of sealer diameter for the joint. Apply generously as the excess will be removed by the shaping spatula. Remove the guide tape and any excess sealer with wax and grease removed and a shop towel.



CAUTION: Use any appropriate protective clothing or equipment when working with chemicals (e.g.: gloves, eye protection, and respiratory protection) and observe all safety procedures in the work place and as required by law.

HINT: Many manufacturers of seam and joint sealers have developed products that can produce factory sealer appearance. Products known as non-sag sealers can be applied with tape affixed to both sides of the area to be sealed. Once the sealer has been applied, the sealer settles in a flattened, broad shape. Once the tape is removed, the edges are straight and clean. This type of sealer is easier to use and helps produce the appearance of factory sealer. Always follow the manufacturer's instructions and safety precautions at all times when working with and applying these products.

Refer to the appropriate Toyota/Scion Collision Repair Manuals for detailed instructions and locations of joints and seams to be sealed. For Toyota or Scion Collision Centre facilities, visit the ETAS and Toyota Technical Information System (TIS) site located in the Infostream website portal at <http://infostream.toyota.ca/> to access Repair Manuals for Collision Damage and other Collision Repair Information Bulletins. For independently operated collision centres interested in these publications, visit www.techinfo.toyota.com for a subscription and access.



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In partnership with Würth Canada, Toyota Canada Inc. (TCI) recommends and offers a complete line-up of body shop supplies, adhesives, fasteners, abrasives, connectors, safety supplies, tools, etc., that are available through the Würth Body Shop Products Program. Toyota and Scion Dealers can order all of Würth's products through this program. Dealers can access the program details in the Infostream website portal at <http://infostream.toyota.ca/>. For independently operated collision centres interested in these products, please contact your local Toyota and Scion Dealer.

In regards to this bulletin, Toyota Canada Inc. recommends the following Würth products:

Würth Product Name	Art. No.	Illustration
Bond + Seal	890.1001	
Pneumatic Application Gun PS 3	891.628	